Deep Learning for AI-Assisted Medical Diagnosis and Automatic Speech Recognition: A Two-Module Study on Chest X-Ray Analysis and CTC-Based Speech-to-Text

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***Abstract*—This work presents the design, implementation, and evaluation of two independent deep learning systems. The first is a computer-vision system for AI-assisted diagnosis of respiratory disease from chest X-rays, combining multi-class classification (Normal, Pneumonia, COVID-19, Tuberculosis) via transfer learning with ResNet-50 and DenseNet-121, and lung segmentation with a U-Net trained on the Montgomery and Shenzhen datasets. The second is an automatic speech recognition (ASR) system based on Connectionist Temporal Classification (CTC) that converts speech to text without frame-level alignment, using a CNN+BiGRU acoustic model. On a held-out test set of 771 images, DenseNet-121 reached an accuracy of 90.9% and a macro F1-score of 0.931, outperforming ResNet-50 (89.9% accuracy, 0.910 macro F1). The U-Net achieved a mean Intersection-over-Union (IoU) of 0.929 and a mean Dice score of 0.962. For ASR, a CNN+BiGRU+CTC model trained on the sentence-level LJ Speech corpus reached a 33.47% Word Error Rate (WER) and an 8.98% Character Error Rate (CER) on its own speaker. A controlled speaker-generalization experiment then evaluated the LJ Speech model on a speaker-disjoint LibriTTS test set of unseen voices, where it degraded to 83.39% WER and 43.05% CER; warm-started fine-tuning on the multi-speaker LibriTTS corpus recovered this to 65.49% WER and 26.34% CER on the same unseen speakers, an absolute reduction of 17.9 WER points. The results confirm that transfer learning and CTC are effective, reproducible building blocks for both diagnostic imaging and speech-to-text under modest, Colab-scale compute budgets, while also exposing the well-known difficulty of open-vocabulary, speaker-independent sentence recognition with a compact, character-level acoustic model.**

***Index Terms*—Deep learning, transfer learning, convolutional neural networks, medical image classification, semantic segmentation, U-Net, Grad-CAM, automatic speech recognition, Connectionist Temporal Classification, word error rate.**

I. Introduction

DEEP learning has become the dominant paradigm across both computer vision and speech processing, displacing hand-engineered pipelines with end-to-end models trained directly on raw or lightly processed data. Two application domains illustrate this shift particularly well: medical image interpretation, where convolutional neural networks (CNNs) now rival expert readers on several screening tasks, and automatic speech recognition (ASR), where alignment-free sequence models have removed the need for hand-aligned phonetic transcripts.

This paper documents a project comprising two independent yet methodologically related modules. The motivation for the first module is clinical: early detection of respiratory diseases such as pneumonia, COVID-19, and tuberculosis is decisive for patient outcomes and for limiting transmission, yet many regions face a shortage of trained radiologists. An accurate, explainable model that can triage chest X-rays offers concrete decision support. The motivation for the second module is interactional: converting speech to text underpins voice assistants, accessibility tools, and automatic transcription, and CTC-based models make this feasible without expensive frame-level annotation.

The contributions of this work are as follows. (i) We build and compare two transfer-learning classifiers (ResNet-50 and DenseNet-121) for four-class chest X-ray diagnosis, addressing class imbalance with a weighted loss and a weighted sampler, and we interpret their decisions with Grad-CAM. (ii) We train a U-Net for lung segmentation on the combined Montgomery and Shenzhen datasets and evaluate it with IoU and Dice. (iii) We implement a complete CNN+BiGRU+CTC ASR pipeline and evaluate it in two settings of increasing difficulty—in-domain, single-speaker sentence recognition and a multi-speaker generalization test—and quantify, on an identical speaker-disjoint test set, how much warm-started fine-tuning improves recognition of voices never seen in training. All code runs in a single GPU-enabled Google Colab notebook and is designed for reproducibility.

The remainder of the paper is organized as follows. Section II reviews the theoretical background. Section III describes the datasets, architectures, and training procedures. Section IV reports and analyzes the results. Section V concludes and outlines future work.

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II. Theoretical Framework

A. Transfer Learning with CNNs

Transfer learning reuses representations learned on a large source dataset (here, ImageNet) and adapts them to a smaller target task. We use two complementary backbones. *ResNet-50* [1] introduces residual (skip) connections that let very deep networks be trained without vanishing gradients, providing a robust, widely validated baseline. *DenseNet-121* [2] connects each layer to every subsequent layer within a block, encouraging feature reuse and parameter efficiency; it is the backbone of CheXNet [6], a reference model for chest X-ray pathology detection. In both cases the convolutional backbone is frozen and only a new classification head is trained, so that the pretrained features are preserved while the model learns the target decision boundary.

B. Semantic Segmentation and U-Net

Segmentation assigns a class to every pixel. The *U-Net* [3] is an encoder–decoder architecture with skip connections that pass high-resolution features from the contracting path to the expanding path, recovering fine spatial detail lost during down-sampling. Its data efficiency makes it the de-facto standard for biomedical segmentation. Performance is measured by the Intersection-over-Union (IoU, or Jaccard index) and the Dice similarity coefficient:

$$\mathrm{IoU} = \frac{|A \cap B|}{|A \cup B|}, \qquad \mathrm{Dice} = \frac{2\,|A \cap B|}{|A| + |B|}, \tag{1}$$

where A is the predicted mask and B the ground-truth mask. Both lie in $[0, 1]$, with higher values indicating better overlap.

C. Model Interpretability: Grad-CAM

Because clinical models must be auditable, we use Gradient-weighted Class Activation Mapping (Grad-CAM) [4]. Grad-CAM weights the activation maps of the final convolutional layer by the gradients of the target class score, producing a heatmap that highlights the image regions most responsible for a prediction. This allows visual verification that the model attends to anatomically plausible regions rather than spurious artifacts.

D. Connectionist Temporal Classification

Classical ASR requires an explicit alignment between acoustic frames and output symbols. CTC [5] removes this requirement by introducing a special *blank* token and marginalizing over all alignments that collapse to the same label sequence. Given a per-frame log-probability distribution y over the vocabulary, the CTC loss maximizes the total probability of all valid alignments π of the target $\mathbf{l}$:

$$\mathcal{L}_{\mathrm{CTC}} = -\log \sum_{\pi \in \mathcal{B}^{-1}(\mathbf{l})} \prod_{t=1}^{T} y^{t}_{\pi_t}, \tag{2}$$

where $\mathcal{B}$ is the collapse operator that removes repeated symbols and blanks. Decoding the trained model with a greedy (argmax-per-frame) rule followed by $\mathcal{B}$ yields the predicted text. ASR quality is reported as Word Error Rate (WER) and Character Error Rate (CER):

$$\mathrm{WER} = \frac{S + D + I}{N}, \tag{3}$$

where S, D, and I are word substitutions, deletions, and insertions, and N is the number of reference words; CER uses the same expression at the character level.

TABLE I
Chest X-Ray Classification Dataset Composition

Class	Train	Val	Test	Total
Normal	1341	8	234	1583
Pneumonia	3875	8	390	4273
COVID-19	460	10	106	576
Tuberculosis	650	12	41	703
Total	**6326**	**38**	**771**	**7135**

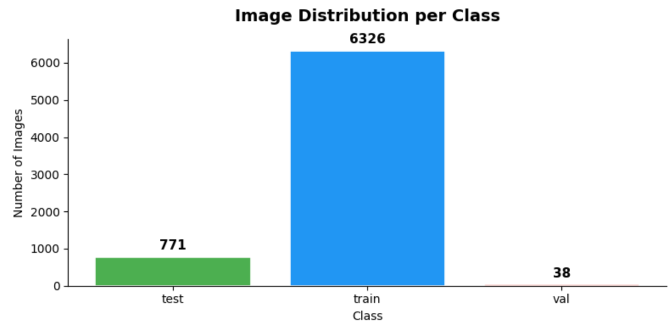

Fig. 1. Image distribution per class for the classification dataset (train folder). The strong dominance of the Pneumonia class motivates the use of a weighted loss and a weighted sampler during training.

III. Methodology

A. Datasets

1) Chest X-Ray Classification: The primary dataset contains 7,135 chest radiographs labeled into four classes. As Table I and Fig. 1 show, the distribution is markedly imbalanced, with Pneumonia dominating and Tuberculosis and COVID-19 underrepresented. The data were partitioned into training (6,326), validation (38), and test (771) subsets following the dataset's predefined folders.

2) Lung Segmentation: For segmentation we combined the *Montgomery County* (138 posteroanterior radiographs with manually annotated left/right lung masks) and *Shenzhen Hospital* (566 usable images with masks) tuberculosis datasets, yielding 704 image–mask pairs. These were split into 492 training, 106 validation, and 106 test pairs. Left and right lung masks were merged by a binary OR into a single binary lung mask.

3) Automatic Speech Recognition: Two audio corpora were used. *LJ Speech* contains 13,100 sentence-level clips (1–10 s) from a single female speaker, with an open vocabulary of 13,821 distinct words, and is used to learn the CTC ASR task in-domain. *LibriTTS* is a multi-speaker corpus; from it we discovered 33,236 utterances across 247 speakers and built *speaker-disjoint* training (173 speakers, 5,000 clips), validation

TABLE II
MODEL PARAMETER COUNTS

Model	Total params	Trainable params
ResNet-50	24,559,172	1,051,140
DenseNet-121	7,480,708	526,852
U-Net	13,601,361	13,601,361
CTC (CNN+BiGRU)	3,334,173	3,334,173

(37 speakers, 1,000 clips), and test (37 speakers, 1,500 clips) splits so that validation and test voices are never seen during training.

B. Preprocessing and Augmentation

All classification images were resized to 224×224 and normalized with ImageNet statistics. Training applied light, diagnosis-preserving augmentation: horizontal flips (p=0.5), small rotations ($\pm 10°$), brightness/ contrast jitter, and mild Gaussian noise; validation and test images received only resizing and normalization. Segmentation images were processed at 512×512 as single-channel (grayscale) inputs.

For ASR, audio was resampled to 22.05 kHz (LibriTTS is natively 24 kHz), padded or truncated to a fixed maximum length, and converted to 40-coefficient Mel-Frequency Cepstral Coefficients (MFCC) with per-utterance mean–variance normalization. Transcriptions were lowercased and reduced to a character vocabulary of 29 tokens (a CTC blank, 26 letters, space, and an unknown token). For the LibriTTS fine-tuning stage we added strong audio augmentation (pitch shift, time stretch, random gain, additive noise) and SpecAugment-style frequency/time masking to improve robustness to unseen voices.

C. Classification Models and Training

Both backbones were initialized with ImageNet weights, their convolutional layers frozen, and their original heads replaced by a two-layer classifier (Dropout $\rightarrow$ Linear(512) $\rightarrow$ ReLU $\rightarrow$ Dropout $\rightarrow$ Linear(4)). Table II summarizes model sizes. Training used the Adam optimizer (learning rate 10^{-4}), a weighted cross-entropy loss together with a `WeightedRandomSampler` to counter class imbalance, automatic mixed precision, a cosine-style learning-rate decay, and early stopping (patience 5) with best-checkpoint restoration. ResNet-50 selected its best model at epoch 12 and DenseNet-121 at epoch 10.

D. Segmentation Model and Training

The U-Net was trained on the merged dataset with a combined Dice/BCE objective, Adam optimization, and early stopping (patience 7), which halted training at epoch 22 with the best model at epoch 15. Validation IoU and Dice were monitored throughout.

E. ASR Model and Training

The acoustic model is a CNN+RNN+CTC network: a three-layer 1-D convolutional encoder (channels $40 \rightarrow 128 \rightarrow 128 \rightarrow 256$, batch-normalized, ReLU) extracts local acoustic patterns, a three-layer bidirectional GRU (hidden size 256) models temporal context, and a linear projection followed by a log-softmax produces per-frame distributions over the 29-token vocabulary. The model is trained with `nn.CTCLoss`, Adam, gradient clipping for RNN stability, and a `ReduceLROnPlateau` scheduler. For LibriTTS we *warm-start* from the LJ Speech checkpoint with a fresh optimizer and a lower learning rate (10^{-4}), adapting the single-speaker model to many speakers without discarding its learned acoustic knowledge. A self-contained inference path also lets the trained model transcribe arbitrary user-supplied `.wav`/`.m4a` recordings without re-running the training pipeline.

TABLE III
AGGREGATE CLASSIFICATION METRICS ON THE TEST SET (771 IMAGES)

Metric	ResNet-50	DenseNet-121
Accuracy	0.8988	**0.9092**
Precision (macro)	0.8914	**0.9268**
Recall (macro)	0.9312	**0.9373**
F1-score (macro)	0.9096	**0.9313**
F1-score (weighted)	0.8991	**0.9097**

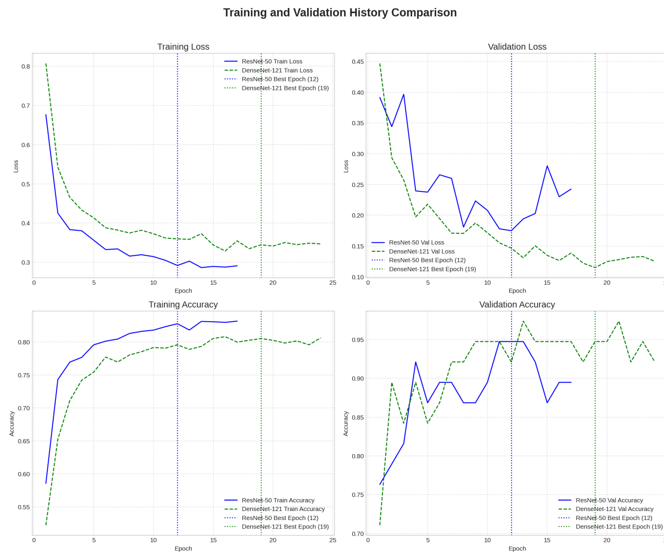

Fig. 2. Training and validation curves for ResNet-50 and DenseNet-121. Dotted vertical lines mark the best epoch selected by early stopping.

IV. RESULTS AND ANALYSIS

A. Classification Performance

Table III reports the aggregate test metrics for both classifiers. DenseNet-121 outperformed ResNet-50 on every aggregate measure, reaching 90.9% accuracy and a 0.931 macro F1-score despite having roughly one third of the parameters, consistent with the parameter-efficiency benefits of dense connectivity. Fig. 2 shows the training dynamics: both models converge smoothly, and DenseNet-121 attains a lower and more stable validation loss.

Per-class results (Table IV, Fig. 3) reveal where the models differ. Both excel on COVID-19 (F1 $\approx$ 0.958), and DenseNet-

TABLE IV
PER-CLASS F1-SCORE (TEST SET)

Class	ResNet-50	DenseNet-121
Normal	0.8470	0.8619
Pneumonia	0.9120	0.9168
COVID-19	0.9581	0.9589
Tuberculosis	0.9213	**0.9877**

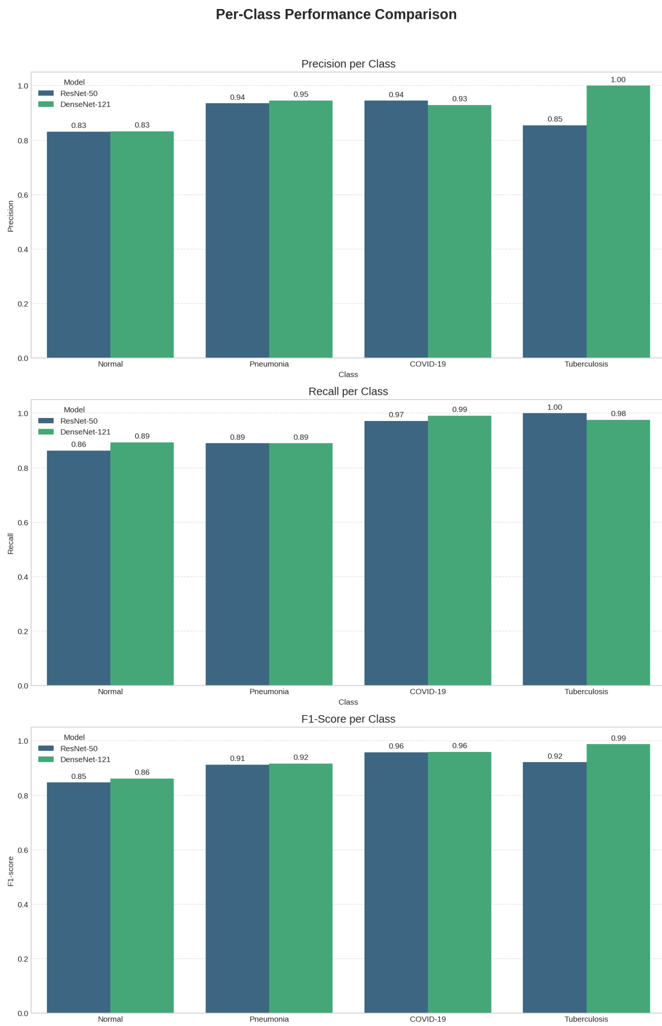

Fig. 3. Per-class precision, recall, and F1-score for both classifiers.

121 is nearly perfect on Tuberculosis (F1 = 0.988, precision 1.000). The hardest class for both models is Normal, which is mutually confused with Pneumonia.

B. Error Analysis

The confusion matrices in Fig. 4 confirm that almost all errors occur between Normal and Pneumonia. For DenseNet-121, 41 Pneumonia cases were predicted as Normal and 20 Normal cases as Pneumonia, while COVID-19 (99.1% recall) and Tuberculosis (97.6% recall) were rarely confused with anything else. This pattern is clinically intuitive: mild or early pneumonia can present with subtle opacities that overlap with normal variation, whereas COVID-19 and advanced

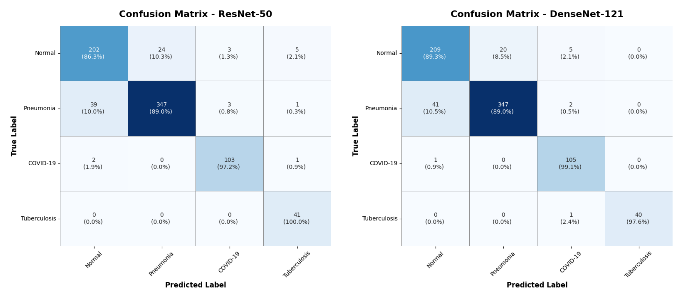

Fig. 4. Confusion matrices on the test set: ResNet-50 (left) and DenseNet-121 (right). Errors concentrate on the Normal/Pneumonia pair.

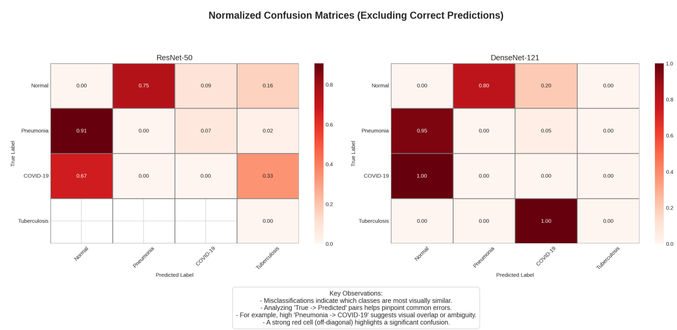

Fig. 5. Normalized confusion matrices with the diagonal removed, isolating the relative error structure of each model.

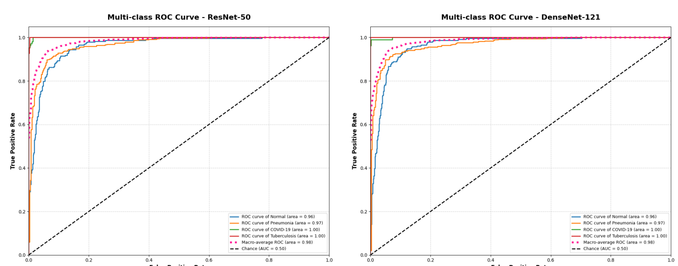

Fig. 6. One-vs-rest ROC curves: ResNet-50 (left) and DenseNet-121 (right).

tuberculosis produce more distinctive radiographic signatures. The normalized error heatmaps in Fig. 5 reinforce that the Normal↔Pneumonia boundary is the dominant source of residual error for both models.

The ROC curves (Fig. 6) show high one-vs-rest separability for every class, with near-unity area under the curve for COVID-19 and Tuberculosis.

C. Interpretability

Grad-CAM visualizations (Fig. 7) indicate that, for correctly classified cases, the models attend to lung-field regions rather than image borders or annotation markers, supporting the clinical plausibility of their decisions. Inspecting the heatmaps of misclassified cases was likewise useful for understanding the Normal/Pneumonia confusion noted above.

D. Segmentation Performance

On the 106-image segmentation test set the U-Net achieved a mean IoU of **0.9290** and a mean Dice score of **0.9625**.

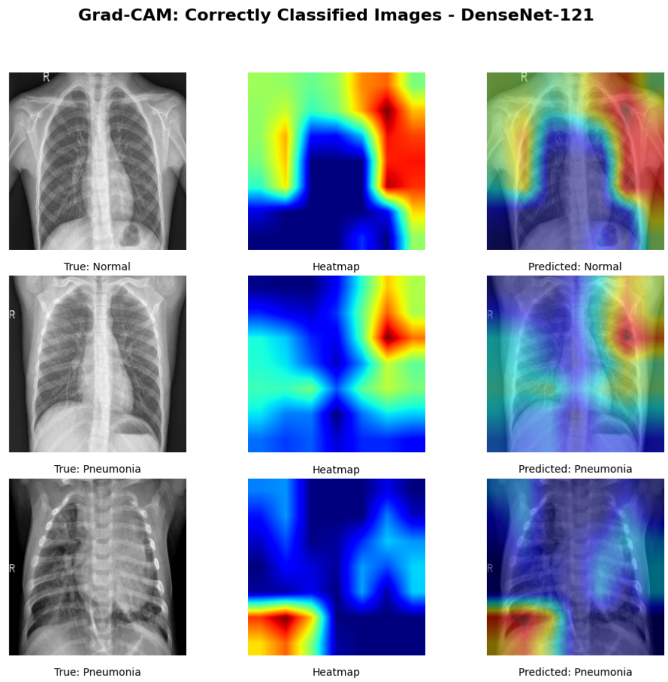

Fig. 7. Representative Grad-CAM heatmaps for correctly classified DenseNet-121 predictions. Warm regions denote areas that most influenced the decision.

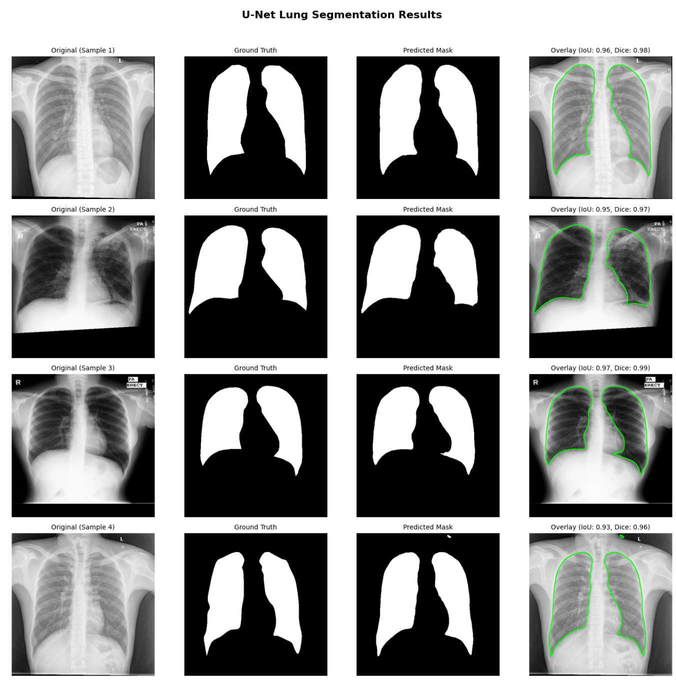

Fig. 9. Qualitative U-Net segmentation results. Columns: input radiograph, ground-truth mask, predicted mask, and predicted-boundary overlay.

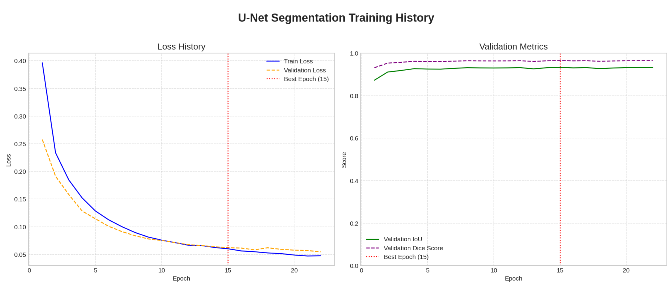

Fig. 8. U-Net training history: loss (left) and validation IoU/Dice (right).

TABLE V
ASR RESULTS ACROSS SETTINGS

Setting (test set)	**WER**	**CER**
LJ Speech (single-speaker sentences)	33.47%	8.98%
LibriTTS (unseen-speaker sentences)†	65.49%	26.34%

† Speaker-disjoint test set; model warm-started from the LJ Speech checkpoint and fine-tuned on multi-speaker LibriTTS data.

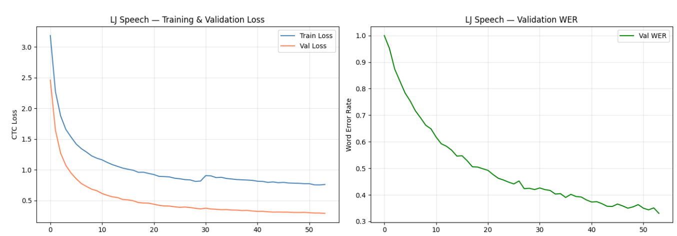

Fig. 10. LJ Speech CTC training history (loss and validation WER).

Because a Dice score above 0.85 is generally regarded as clinically acceptable for lung delineation, these values indicate accurate and reliable segmentation. The training history (Fig. 8) shows rapid convergence, with validation Dice exceeding 0.93 within the first few epochs. Qualitative results (Fig. 9) show predicted masks that closely track the ground-truth lung boundaries, with per-sample Dice values of 0.96–0.99.

E. Speech Recognition Performance

The ASR results, summarized in Table V, span two settings of increasing difficulty: in-domain recognition on the model's own speaker, and generalization to unseen speakers. On open-vocabulary, single-speaker sentences (LJ Speech) the CTC model reached a WER of 33.47% while the CER remained low at 8.98%. This large WER–CER gap is characteristic of a compact character-level model: it produces near-phonetic spellings (e.g., *"semitary"* for *"cemetery"* and *"uter"* for *"utter"*) that are mostly correct character by character but counted as whole-word errors, since the model has no lexicon or language model to enforce valid spellings. The training and validation curves are shown in Fig. 10.

The hardest setting is generalization to *unseen speakers*. To measure this honestly, we took the LJ Speech model unchanged and evaluated it on the speaker-disjoint LibriTTS test set—voices it was never trained on. Its performance collapsed from the 33.47% in-domain WER to 83.39% WER (43.05% CER), confirming that a model trained on a single voice does not transfer to arbitrary speakers. We then warm-started from that same checkpoint and fine-tuned on the multi-speaker LibriTTS corpus. Evaluated on the *identical* unseen-speaker test clips, the fine-tuned model recovered to 65.49% WER and 26.34% CER—an absolute reduction of 17.9 WER points and 16.7 CER points purely from exposure to speaker diversity. Table VI isolates this before/after comparison, and Fig. 11 shows the corresponding fine-tuning curve.

TABLE VI
CROSS-DOMAIN GENERALIZATION ON THE IDENTICAL UNSEEN-SPEAKER TEST SET

Model (evaluated on unseen LibriTTS speakers)	WER	CER
LJ Speech model (before fine-tuning)	83.39%	43.05%
LibriTTS-fine-tuned model (after)	**65.49%**	**26.34%**
Absolute improvement	**17.90 pts**	**16.71 pts**

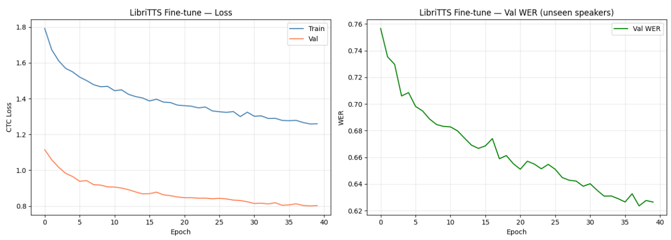

Fig. 11. LibriTTS warm-start fine-tuning: training loss and validation WER on unseen speakers, which decreases steadily as the model adapts to speaker diversity.

F. Discussion

Across both modules the results are coherent and reproducible. For imaging, transfer learning with a parameter-efficient backbone (DenseNet-121) delivered the strongest classification, and U-Net produced clinically usable lung masks; together these form a plausible pipeline in which segmentation supplies a region of interest for downstream classification. For speech, CTC proved excellent for constrained command recognition and serviceable at the character level for open speech, while the speaker-generalization experiment honestly exposed both the limits and the adaptability of a small, language-model-free system: a single-speaker model transfers poorly to new voices (83.39% WER), yet warm-started fine-tuning on diverse speakers recovers a large fraction of that loss (down to 65.49% WER) without any architectural change. The persistently high absolute WER, together with the much lower CER, indicates that the dominant remaining error is the absence of a lexicon or language model rather than overfitting to one voice. The principal threat to the imaging results is the very small validation set (38 images), which makes per-epoch validation metrics noisy; the test set (771 images), however, is sufficiently large for the reported comparisons to be meaningful.

V. CONCLUSIONS AND FUTURE WORK

We presented two independent deep learning systems and evaluated them end-to-end. In the computer-vision module, DenseNet-121 achieved 90.9% accuracy and a 0.931 macro F1-score for four-class chest X-ray diagnosis, surpassing ResNet-50, and a U-Net reached 0.929 mean IoU and 0.962 mean Dice for lung segmentation. In the speech module, a CNN+BiGRU+CTC model obtained 33.47% WER and 8.98% CER on in-domain single-speaker sentences; a controlled generalization test showed the single-speaker model degrading to 83.39% WER on unseen voices, and a warm-started fine-tune on multi-speaker LibriTTS data recovered this to 65.49% WER (26.34% CER) on the same unseen speakers. Grad-CAM confirmed that the classifiers attend to anatomically relevant regions, addressing the explainability requirement of clinical decision support.

Several directions would strengthen the work. For classification, unfreezing and fine-tuning the upper backbone layers, expanding the validation set, and using the U-Net masks to crop lung regions before classification could raise accuracy and reduce the Normal/Pneumonia confusion. For segmentation, a stronger encoder (e.g., an EfficientNet backbone) and test-time augmentation could push Dice higher. For ASR, the largest gains would come from integrating a language model or beam-search decoding with a lexicon, increasing the training budget beyond the Colab-limited 5,000-clip subsets, and adopting a Transformer or Conformer acoustic model. Combining the modules into a single real-time assistant—transcribing a clinician's spoken notes while analyzing an X-ray—remains an appealing integration goal.

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